Analyzing the Influence of Neighbourhood and Other Property Attributes on Calgary's 2025 Property Assessment Values

Group Members: Bhuvan Dubey (30139765), Jimoh Adewumi (30295456), Adewale Adewumi (30226120)

INTRODUCTION

Motivation

This project looks at factors on a parcel level that influence the assessed values of properties using Calgary's municipal assessment data. We want to find out what features of parcels and neighbourhoods explain the changes in assessed property values in the current assessment year. Municipal assessments are influenced by market dynamics as well as policy decisions. Knowing what causes assessed values to change is helpful to stakeholders in identifying and addressing issues of equity, spotting anomalies, and determining the priorities for infrastructure and development. We, as a team, are committed to turning parcel-level data into insights that can be used by a diverse range of stakeholders, analysts, urban planners, real estate professionals, researchers, residents, and potential homebuyers. Our goal is to reveal the hidden value in property data to facilitate smarter decisions and to provide a more complete view of Calgary's housing market. The group members have the background and experience with economic and administrative data and are enthusiastic about the idea of converting parcel-level records into actionable insights. In doing so, city analysts, planners, real estate people, researchers, and residents will be benefited as they will be able to see the patterns and it will become clearer why property values change from place to place.

Background

Property assessments are required for the allocation of property taxes by municipalities, and they are meant to have a relationship to the activity in the local real estate market. Property assessments are based on, but not limited to, parcel size, land use designation, location, characteristics of the building (age and building type). The ability to compare assessments between communities facilitates an assessment of socio-economic and development differences of neighborhoods through property taxation across the city. Statistical controls can also be applied to explore parcel level effects versus community wide effects. Many previous studies have identified neighborhood location as one of the strongest predictors of housing value, potentially linked to proximity to services, amenities and employment (Gyourko et al., 2006). However, parcel size and built form remain important contributors to value within communities. By modelling these together, research will be able to shed light on whether property tax differences represent actual differences in market fundamentals versus equity issues in the assessment process.

Data Background

Our population of interest is all taxable parcels in the City of Calgary's Current Year Property Assessments dataset, including all communities and property types. We will analyze key parcel level variables including assessed value, property type, land size, land use designation, and community location. All information was collected and compiled by the City of

Calgary's Assessment & Tax Department as a part of the annual property assessment process. Characteristics for each parcel, which include land area, legal boundaries, and land use designation, were accessed in municipal land registry and planning documents. Characteristics of every building (including construction year, property type, and floor area) were derived through a combination of building permit records, field inspections, and verified owner submissions. Assessed values (NR_ASSESSED_VALUE, FL_ASSESSED_VALUE, and total ASSESSED_VALUE) were generated through mass appraisal methodologies using baseline market data from neighboring comparable property sales that were also subjected to review. All data are maintained within the municipal assessment database and provided in open-data format (where no personally identifiable information is provided). The data is publicly available and licensed for academic use. The assessment period is 2025 and historical data is also available which might be useful for future analysis. For more details about the variables, please see Appendix Table 1.

Guiding Question

What are the key parcel-level and community-level factors that influence assessed property values and assessed value per square foot in Calgary?

<u>METHODS</u>

Question 1: Which communities show the highest average assessed values and lowest average assessed values?

Background/Context: The assessed values of a property help to determine the value of the parcel for taxation. Property assessment values can greatly differ based on several variables such as the use of land, the area of the parcel, where it is located, and type of property, etc. Certain types of communities, particularly those containing a large number of larger properties or institutional type properties, tend to have an average assessment value that is significantly higher compared to other community types, and those that contain more evenly mixed types of assessed values or have modest type assessment values. By evaluating the average assessment values across the city, we can identify which communities/neighbourhoods have the most value on average, and identify any potential inequities or patterns amongst the value placed on properties in the city.

Modelling Workflow: Initially, we loaded our data from the data source and proceeded to clean it by changing the assessed values into figures and removing those parcels which had no data or had wrong data. After that, we selected and calculated a set of simple statistics for each community, these involved average assessed value, median assessed value, standard deviation, and interquartile range (IQR). The data we had allowed us to figure out the communities with the highest and lowest average assessed property value and also the widest spread or variability among the assessed property values which is a sign of possible inequities. At the end of the day, we created tables and bar charts to compare the communities and came up with the interpretation of the results in relation to land use and neighborhood characteristics.

Question 2: How does assessed value per square foot vary across land use designations and communities?

Background/Context: Assessed value per square foot gives a better idea of property density and valuation relative to parcel size. Different land use designations (like R-G, R-CG, R-Gm) usually have different typical values per square foot. Also, within communities, some parcels might be much more valuable than others, which may suggest land-use mix or socio-economic differences. Analysis of these differences helps us understand where high-value or diverse land-use areas exist, and if certain zones tend to be more expensive per unit area.

R-CG is a residential designation that is primarily for rowhouses but also allows for single detached, side-by-side and duplex homes that may include a secondary suite. The R-G district is for a mix of low density housing forms in suburban greenfield locations, including single-detached, side by side, duplex, cottage housing clusters and rowhouse development, all of which may include secondary suite. R-MH is a residential designation that is for mobile home parks. The R-Gm district is for a range of low density housing forms in suburban greenfield locations, including side by side, duplex, cottage housing clusters and rowhouse development (but excluding single detached), all of which may include a secondary suite.

Modelling Workflow: We started by importing and cleaning the data—checking assessment values and land use categories, removing missing or bad data. Then, we calculated the average assessed value per square foot for each community and land use zone. In order to analyze the relationships between land use, community, and assessed value per square foot, we built regression models with log-transformed assessed value per square foot as the outcome variable. Model diagnostics, such as residual plots and QQ plots, were used to evaluate if the model assumptions held. Residuals (Figures 7 and 8) were examined to assess model fit—residual plots showed no major pattern, but QQ plots indicated some non-normality, which was partly addressed by the log transformation. We interpreted the results by comparing the mean values and dispersion across land uses and neighborhoods.

Division of Work

We divided the project tasks based on each member's strengths in statistical analysis. Bhuvan focused on Research Question 1, exploring differences in assessed property values across Calgary communities. Adewale worked on Research Question 2, analyzing how assessed value per square foot varies across land-use designations and communities. Jimoh handled the core statistical modeling, building and refining multiple linear regression models, checking assumptions such as linearity, multicollinearity, and normality, and performing hypothesis tests. While the tasks varied in scope, each role was critical to the overall analysis, and together they ensured a balanced, collaborative approach that produced strong, data-driven insights.

STATISTICAL METHODS AND RESULTS

Model

To answer the research questions on the determinants of value, multiple linear regressions are used to model the total assessed value (ASSESSED_VALUE) and assessed value per square metre (VALUE_PER_SQM). The modelling process involves an OLS baseline model, followed by stepwise (forward and backward) and backward elimination using AIC and BIC to select the most parsimonious sets of predictors. Interaction terms (PROPERTY_TYPE × LAND_USE_DESIGNATION) are added to account for effect modification between the parcels' characteristics. Categorical variables such as land use designation and community fixed effects were included as predictors in the regression models. Model diagnostics such as residual plot, multicollinearity check using VIF, and model assumptions are performed to ensure model reliability. Assumption checks included evaluating linearity, homoscedasticity, and independence of residuals to ensure model validity. Multicollinearity was assessed via VIF, and high VIF values indicated severe multicollinearity among some predictors, which was considered in interpretation. Log transform addresses skew and stabilizes variance. Transformations such as the log were chosen based on residual diagnostics indicating non-normality and heteroscedasticity. Overall, the performance of the models is evaluated using the adjusted R^2 and residual standard error, with model selection focused on achieving a balance between interpretability and explanatory power, guided by AIC and BIC criteria.

Rationale

Multiple linear regression fits well with this analysis since the two outcomes of interest, ASSESSED_VALUE and VALUE_PER_SQM, are both continuous variables, and the objective is to quantify the contribution of multiple parcel-level and community-level characteristics in explaining variation in property values. This final modelling approach provides the flexibility to investigate several predictors simultaneously while controlling for confounders like land use designation and location of the community. Also, regression offers some interpretable coefficients that show the direction and magnitude of association of each factor with assessed value, making it an effective technique of understanding the drivers of property valuation in Calgary. Additionally, model diagnostics such as residual analysis and checks for multicollinearity are important to ensure the validity and reliability of the regression results.

RESULTS

Table 2. Model Comparison and Fit

Model	Outcome	Key Predictors	Adjusted R^2	Notes
Full	ASSESSED_VALUE	LAND_SIZE_SM, LAND_SIZE_SF, LAND_SIZE_AC, YEAR_OF_CONSTRUCTION, LAND_USE_DESIGNATION, PROPERTY_TYPE, COMM_NAME	0.5171	LAND_SIZE_SM significant but high VIF; LAND_SIZE_AC and PROPERTY_TYPE not significant

Reduced	ASSESSED_VALUE	LAND_SIZE_SF, YEAR_OF_CONSTRUCTION, LAND_USE_DESIGNATION, COMM_NAME	0.5566	Most covariates significant; improved parsimony
Log-reduced	log(ASSESSED_VALUE)	LAND_SIZE_SF, YEAR_OF_CONSTRUCTION, LAND_USE_DESIGNATION, COMM_NAME	0.5571	Log transform explored; similar fit

Table 3. Selected Significance Tests and Diagnostics

Term / Test	Estimate/Statistic	p-value / Result	Interpretation
LAND_SIZE_AC (levels model)	—	0.065944	Not significant at α=0.05
PROPERTY_TYPE (levels model)	—	0.800345	Not significant; excluded
LAND_USE_DESIGNATION (overall)	—	< 0.05	Reject H0: differences across designations
COMM_NAME (overall)	—	< 0.05	Reject H0: differences across communities
Kolmogorov–Smirnov (residuals)	Not normal	—	Motivated log-transform and robust checks

Table 4. Variance Inflation Factors

Predictor	VIF	Comment
LAND_SIZE_SM	4.197213e+08	Severe multicollinearity
LAND_SIZE_SF	4.197227e+08	Severe multicollinearity
YEAR_OF_CONSTRUCTION	3.3469	Acceptable
LAND_USE_DESIGNATION: R-CG	72.0878	High
LAND_USE_DESIGNATION: R-G	17.114	High
LAND_USE_DESIGNATION: R-Gm	1.9781	Low
LAND_USE_DESIGNATION: R-MH	1.2706	Low

Question 1: Which communities show the highest average assessed values and lowest average assessed values?

Statistical Hypotheses

Null Hypothesis : No difference in mean assessed values across communities.
Alternative Hypothesis : At least one community differs in mean assessed value.

Statistical Analysis

Figures 1 and 2 illustrate the differences in assessed values among communities. Figure 1 ranks the first ten communities in terms of average assessed value. Areas with high incomes such as Bel-Aire, Britannia, and Elbow Park are found to have the highest values. Figure 2 demonstrates the lowest assessed-value Calgary communities and is mainly composed of the southeastern areas (Ward 12 sub-area, Forest Lawn, Erin Woods). The community fixed effects in the regression model (Table 2) are in line with these findings, showing statistically different results ($p < 0.05$), which implies that assessed values vary significantly from one community to another.

Results Interpretation

The analysis shows that wealthy communities such as Bel-Aire and Britannia have the highest average assessed values. Poorer communities in southeast Calgary, like Forest Lawn and Erin Woods, have the lowest. The model's adjusted R^2 of around 0.556–0.557 shows a reasonable fit, keeping the model easy to understand while still explaining a good portion of the variance. These results confirm what we thought: there are socio-economic inequalities between Calgary neighborhoods. Interestingly, some of the newer communities (e.g., Ricardo Ranch) have reasonably high-assessed values which may reflect ongoing developments in the area that may increase property values.

Evaluation of Approach

The regression method, using community fixed effects, is good at capturing differences between neighborhoods. The QQ plot shows that there is some non-normality left over, and there is multicollinearity between size factors. Spatial models or changing the variables could fix these problems. Later studies might use spatial regression or add things like amenities or recent sales data to improve how well we can predict values.

Question 2: How does assessed value per square foot vary across land use designations and communities?

Statistical Hypotheses

Null Hypothesis : No difference in assessed value per square foot across communities.
Alternative Hypothesis : At least one community differs in assessed value per square foot.

Statistical Analysis

The differences in value that were measured per square foot for each of the communities and the types of land use are represented visually through the figures 4, 5, and 6. Figure 4 puts in contrast the top 10 communities by the assessed value per square foot and, among them, places communities such as Currie Barracks and Lower Mount Royal, which feature the highest values. Figure 6 illustrates the typical assessed value per square foot by land use designation, pointing out that R-G (residential general) and R-Gm (medium density) have higher values than R-CG (residential community/government). The residual diagnostics (Figures 7 and 8) show that the data somewhat deviates from normal and hence log-transformations were applied, resulting in better model robustness.

Results Interpretation

This data shows that there are great fluctuations in the assessed values per square foot of different communities and land use designations. It is apparent that such areas as Currie Barracks and Lower Mount Royal have the highest values per square foot which goes in line with their status as wealthy neighborhoods. As for the land use categories, R-G and R-Gm usually have higher assessed values per square foot than R-CG which is a reflection of the differences in density and types of properties. The residual plots hint that even though the log transformation enhances the model fit, some heteroscedasticity is still present which means that using spatial models for further refinement could give more accurate results.

Evaluation of Approach

The visuals for the diagnosis and the log-transformations that were combined in this work have been successful in dealing with the problems of skewness and heteroscedasticity that the data presented. Still, the non-normality of the residuals indicates that the use of other modeling methods, including spatial regression or machine learning algorithms, might more effectively unravel intricate patterns. The inclusion of further variables such as amenities, recent sales, or neighborhood socio-economic factors may also lead to a greater explanatory power.

CONCLUSION

The final work utilized a multiple linear regression model to delve into the aspects that influence the assessed property values in Calgary. The model contained the main predictors such as land size, year of construction, land use designation, and community fixed effects. To remedy skewness and heteroscedasticity, thus improving the robustness of the model, log transformation of the assessed value per square foot was performed. The impacts of assessed value differences between residential neighborhoods and across the land use categories are confirmed by the data, with the greatest values being those in wealthy communities and areas with a higher land use density. The diagnostic tests discovered some residual non-normality and multicollinearity among size variables, thus pointing to the possibilities of further refinement. The model, on the whole, accounted for about 55.7% of the variation in assessed value per square foot and therefore, it could serve as a tool for understanding the major factors that influence property valuation in Calgary.

Future Direction

This analysis establishes a strong link between neighborhood and land use-based disparities in assessed property values and values per square foot in Calgary, with socio-economic factors, development patterns, and land zoning acting as the main drivers. Further research should look at spatial econometric models (e.g., GWR, spatial lag/error models), use historical assessment trends, and compare assessed values to actual market sales for better valuation accuracy and fairness analysis. Another area of interest might be the relationship between over- and under-assessment and neighborhood demographics (an equity analysis); the use of machine-learning models for more accurate predictions; and introducing other covariates such as the number of community amenities and average prices of recently sold properties to name a few for a more holistic understanding of valuation drivers.

REFERENCES

City of Calgary — Current Year Property Assessments (Parcel), Open Calgary.
City of Calgary — 2025 Property Assessment Market Report (market date July 1, 2024; condition date Dec 31, 2024).
City of Calgary — Land Use Bylaw 1P2007 (R-CG, R-G/R-Gm, R-MH definitions).
Gyourko, J., Mayer, C., & Sinai, T. (2006/2013). Superstar Cities. NBER Working Paper 12355; AEJ: Economic Policy.

APPENDIX

Table 1. Variable of Interest

Variable	Type	Definition	Units / Scale	Notes
ASSESSED_VALUE	Continuous	Total assessed value of the parcel/building	CAD	Outcome variable
ASSESSED_VALUE_PSFT	Continuous	Assessed value per square foot	CAD/sq ft	Derived outcome used in visualizations
LAND_SIZE_SF	Continuous	Land area (square feet)	sq ft	Primary land size used in final model
LAND_SIZE_SM	Continuous	Land area (square metres)	sq m	Removed due to multicollinearity
LAND_SIZE_AC	Continuous	Land area (acres)	acres	Not significant at α=0.05
YEAR_OF_CONSTRUCTION	Continuous	Year the primary structure was built	Year	Proxy for building age
LAND_USE_DESIGNATION	Categorical	Zoning district (R-CG, R-G, R-Gm, R-MH, etc.)	Factor	Regulatory land-use category
PROPERTY_TYPE	Categorical	Broad property class	Factor	Excluded (residential-only scope; not significant)
COMM_NAME	Categorical	Community/neighbourhood name	Factor	Location fixed effects

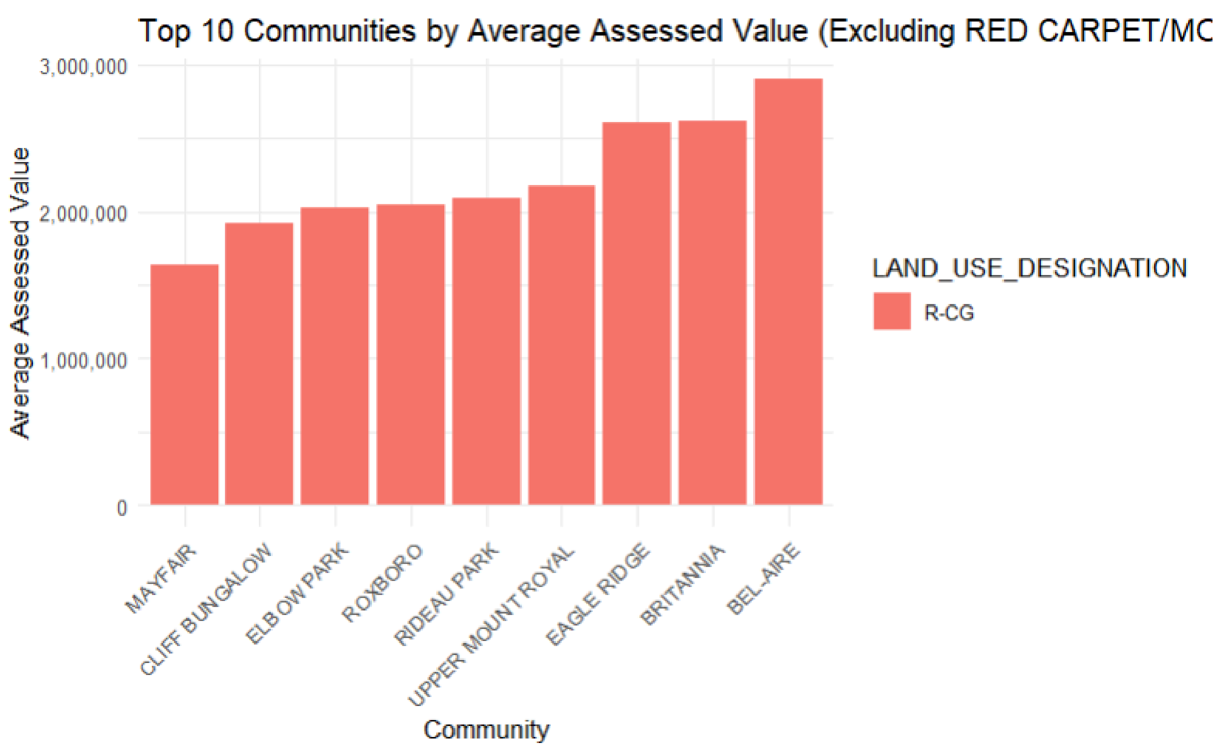

Figure 1. Top 10 Communities by Average Assessed Value (with land-use excluding R-MH)

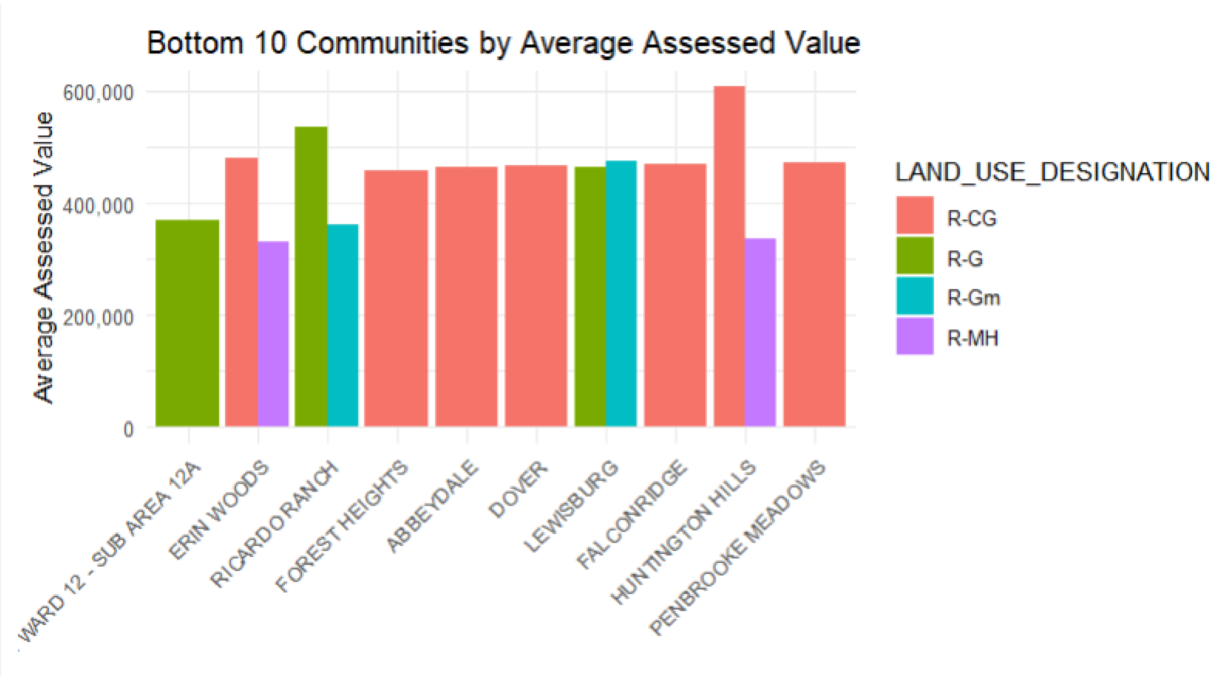

Figure 2. Bottom 10 Communities by Average Assessed Value (with land-use)

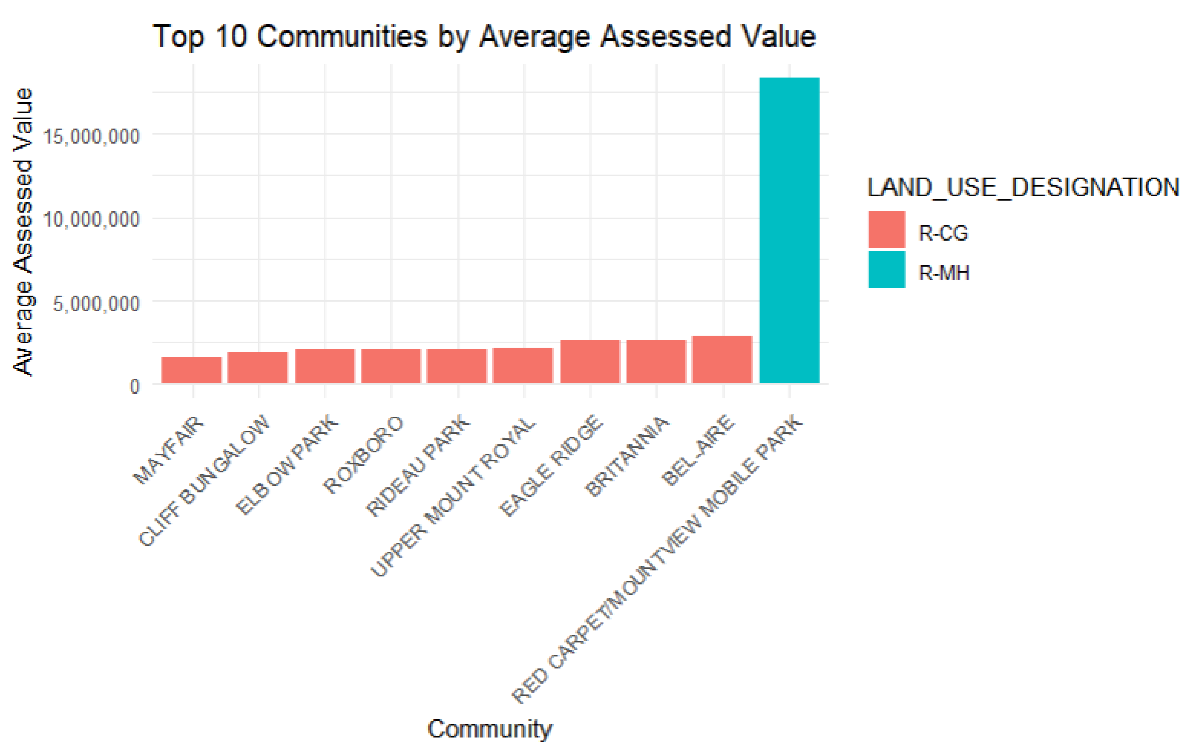

Figure 3. Top 10 Communities by Average Assessed Value (with land-use)

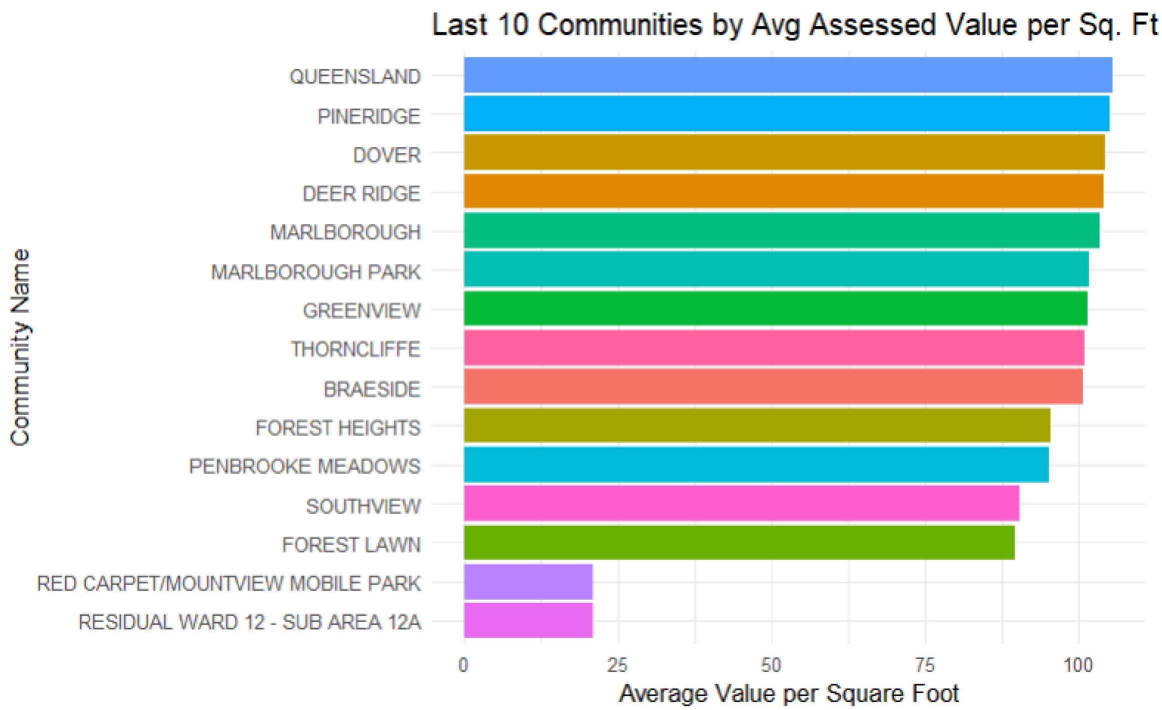

Figure 4. Last 10 Communities By Average Assessed Value per Square Foot across Communities

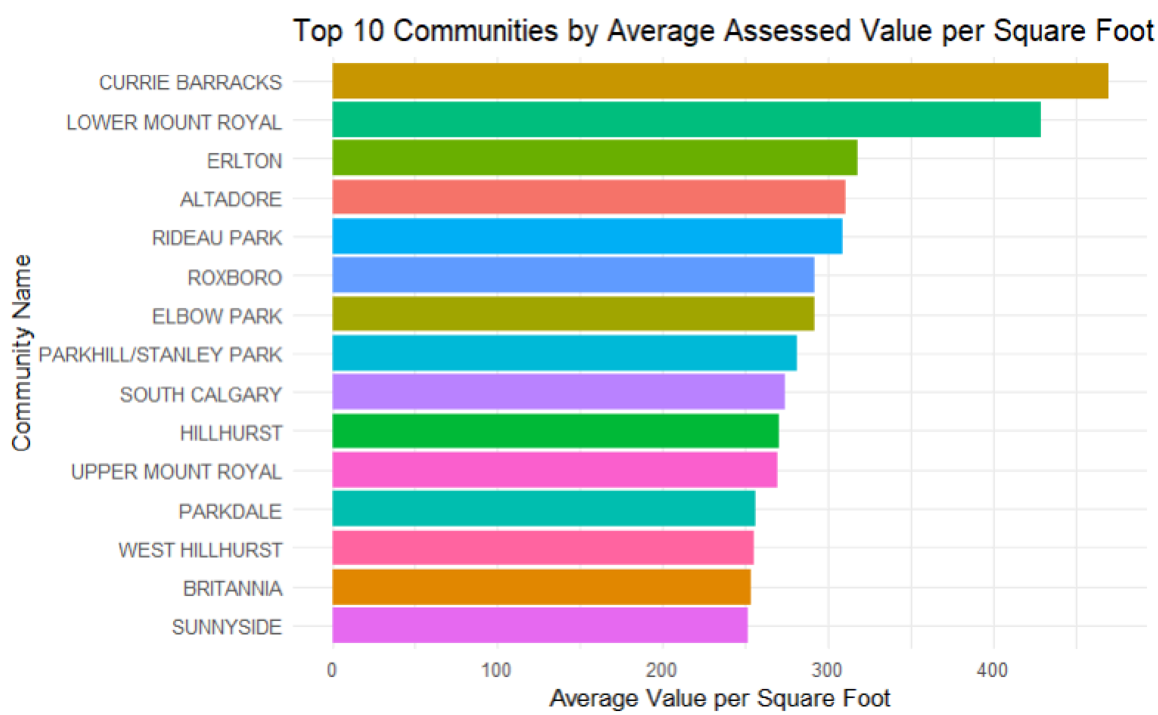

Figure 5. Top 10 Communities By Average Assessed Value per Square Foot across Communities

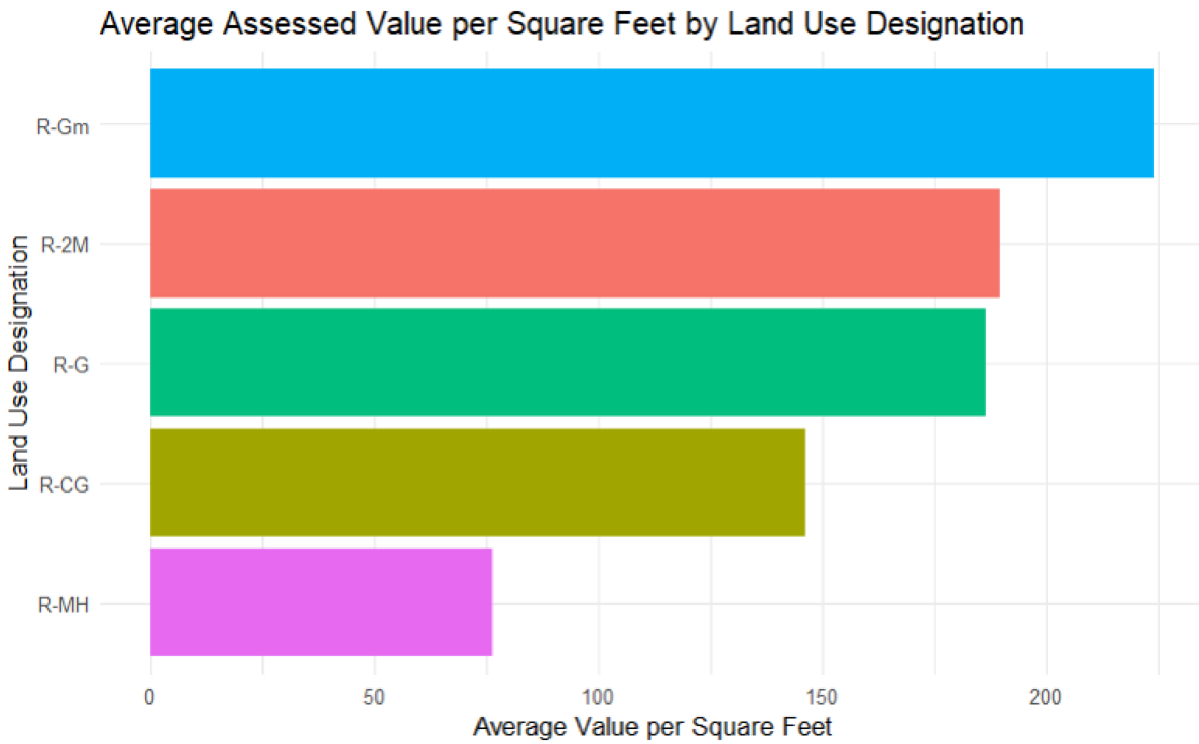

Figure 6. Assessed Value per Square Foot by Land-Use Designation

Model Diagnostics:

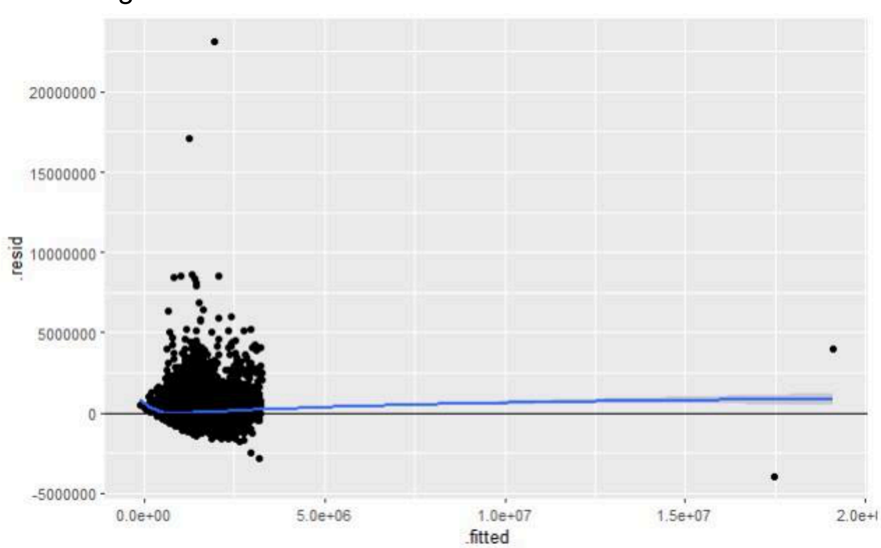

Figure 7. Residual Plot for Regression Diagnostics

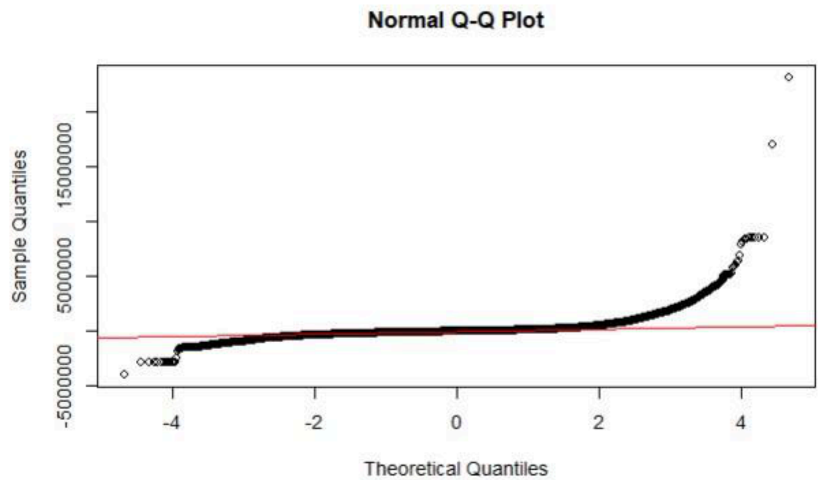

Figure 8. QQ Plot for Normality Check